



An extreme vertices mixture design approach to the optimization of methylal production process using p-toluenesulfonic acid as catalyst

The purpose of this research paper is to investigate and optimize the methylal production process using p-toluenesulfonic acid (PTSA) as catalyst through an Extreme Vertices mixture design.

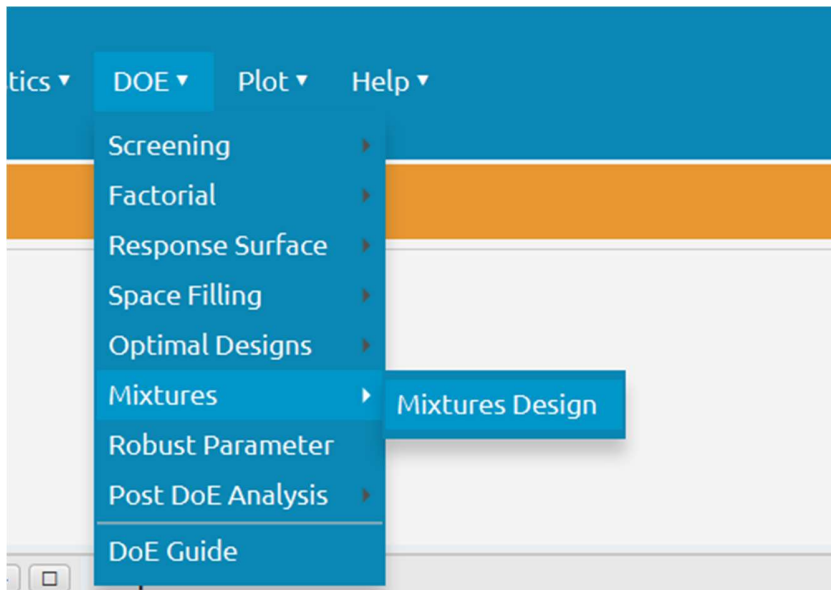
The factors (mixture components) examined are: X_1 = methanol, X_2 = formalin solution and X_3 = PTSA catalyst. The responses examined are: Y_1 = methylal production capacity (mL/100 mL of reactants) and Y_2 = methylal synthesis rate (mL/h). The applied DoE method is Extreme Vertices mixture design.

Isalos version used: 2.2.0

Scientific Article : Damiri, S., Pouretedal, H. R., & Bakhshi, O. (2016). *An extreme vertices mixture design approach to the optimization of methylal production process using p-toluenesulfonic acid as catalyst*. Chemical Engineering Research and Design, 112, 155–162.

Step 1: Extreme Vertices Design

In the first tab named “Action”, rename it to “Extreme Vertices”. Then, go to DOE → Mixtures → Mixtures Design. Select Extreme Vertices, set the number of components to 3, and define the component bounds as $A = 0.40\text{--}0.75$, $B = 0.20\text{--}0.55$, and $C = 0.01\text{--}0.08$. Select Axial Points and Center Points, and click Execute to generate the design.



EW.iap

DoE Mixtures

Number of Replicates: 1

Number of Blocks: 1

☐ Random Standard order

Select Design: Extreme Vertices

Number of components in the Mixture: 3

Degree of Design: 1

Linear Constraints

Augment the Design with Axial Points ☒

Augment the Design with Center Points ☒

Component #	Min	Select Values	Max
A	0.4	0.4	0.75
B	0.2	0.2	0.55
C	0.01	0.01	0.08

Execute Cancel

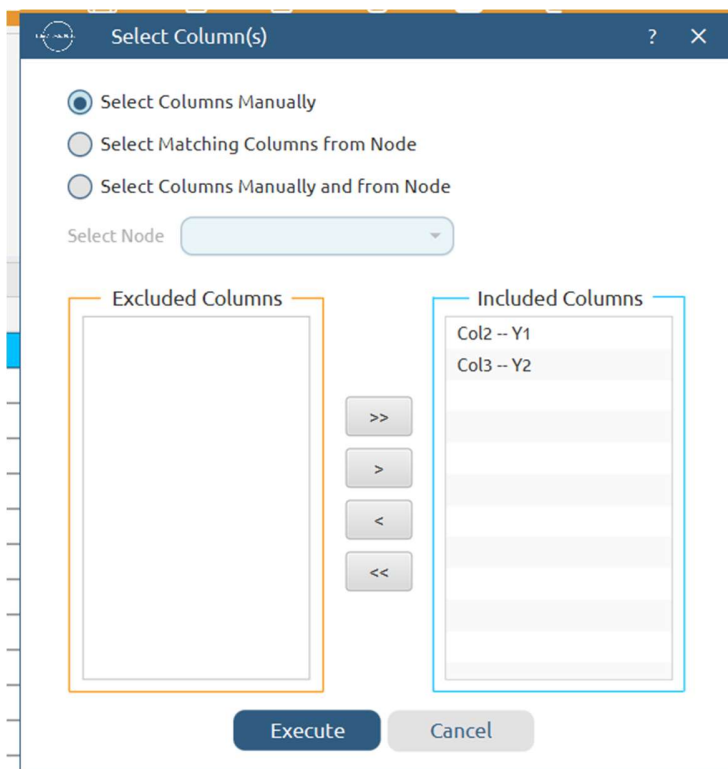
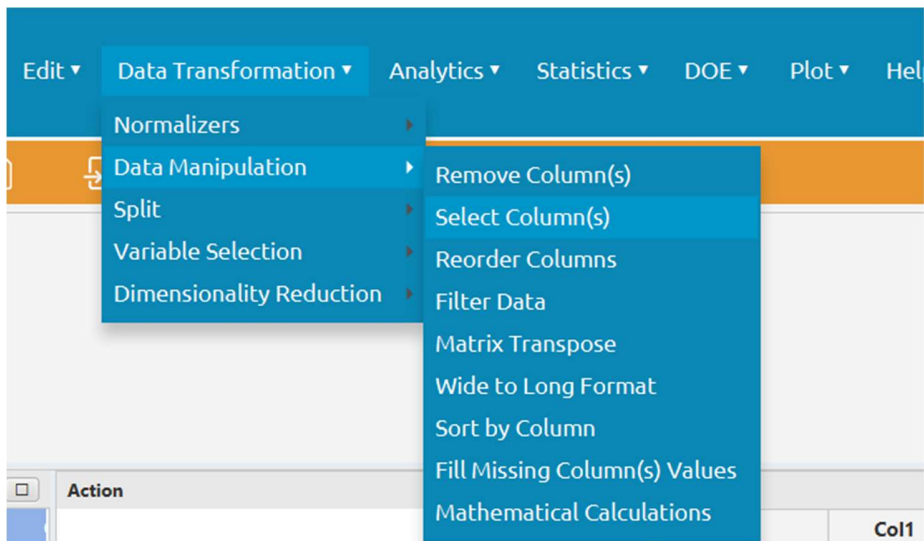
Results (right spreadsheet):

Output									
	Col1	Col2 (I)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (D)	Col7 (D)	Col8 (D)	
User Header	User Row ID	Standard Order	Block Number	Replicate Number	Point Type	A	B	C	
1		1	Block: 1	Replicate: 1	Design Point	0.4	0.55	0.0500000	
2		2	Block: 1	Replicate: 1	Design Point	0.4400000	0.55	0.01	
3		3	Block: 1	Replicate: 1	Design Point	0.4	0.52	0.08	
4		4	Block: 1	Replicate: 1	Design Point	0.75	0.24	0.01	
5		5	Block: 1	Replicate: 1	Design Point	0.75	0.2	0.0500000	
6		6	Block: 1	Replicate: 1	Design Point	0.72	0.2	0.08	
7		7	Block: 1	Replicate: 1	Design Point	0.5766667	0.3766667	0.0466667	
8		8	Block: 1	Replicate: 1	Design Point	0.4883333	0.4633333	0.0483333	
9		9	Block: 1	Replicate: 1	Design Point	0.5083333	0.4633333	0.0283333	
10		10	Block: 1	Replicate: 1	Design Point	0.4883333	0.4483333	0.0633333	
11		11	Block: 1	Replicate: 1	Design Point	0.6633333	0.3083333	0.0283333	
12		12	Block: 1	Replicate: 1	Design Point	0.6633333	0.2883333	0.0483333	
13		13	Block: 1	Replicate: 1	Design Point	0.6483333	0.2883333	0.0633333	

Step 2: Definition of Response Variables

Create a new tab named “Responses”. Define two response columns, Y1 and Y2, and enter the experimental values corresponding to the generated design order. Apply Data Transformation → Select Column(s) and include both response columns.

Input			
	Col1	Col2 (D)	Col3 (D)
User Header	User Row ID	Y1	Y2
1		28.73	17.24
2		20.87	12.4
3		45.12	21.12
4		20.0	12.63
5		24.26	7.32
6		39.33	12.76
7		40.36	13.33
8		25.23	15.93
9		35.46	19.88
10		34.22	21.34
11		40.99	21.58
12		30.55	11.56
13		39.85	24.67
14			



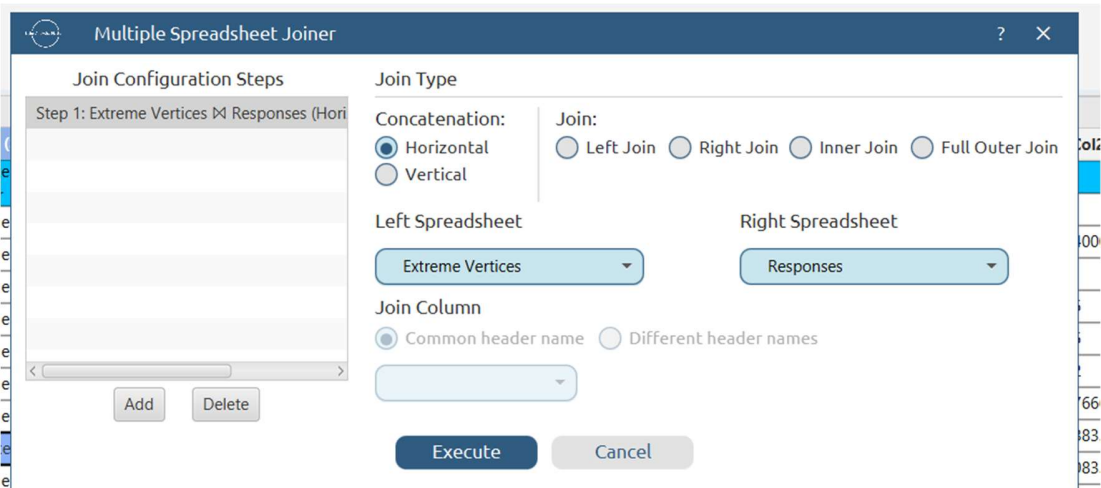
Results :

Output			
	Col1	Col2 (D)	Col3 (D)
User Header	User Row ID	Y1	Y2
1		28.73	17.24
2		20.87	12.4
3		45.12	21.12
4		20.0	12.63
5		24.26	7.32
6		39.33	12.76
7		40.36	13.33
8		25.23	15.93
9		35.46	19.88
10		34.22	21.34
11		40.99	21.58
12		30.55	11.56
13		39.85	24.67

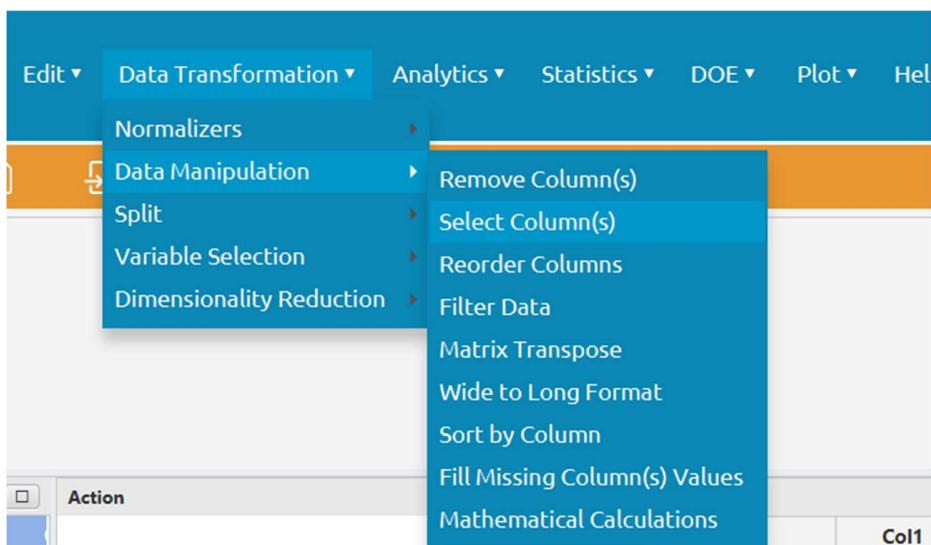
Step 3: Data Selection

Create a tab named “Data”. To combine the mixture design with the response data, right-click on the spreadsheet and select Import from Multiple Spreadsheets. In the Multiple Spreadsheet Joiner, select Horizontal concatenation, with Extreme Vertices as the Left Spreadsheet and Responses as the Right Spreadsheet. Click Execute to combine the two spreadsheets into a single dataset.

In Data Transformation > Data Manipulation > Select Column(s), excluding Standard Order, Block Number, Replicate Number and Point Type. Include only A, B, C, Y1 and Y2. The output of this step must therefore contain exactly five columns: A, B, C, Y1 and Y2.



Input										
	Col1	Col2 (I)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)
User Header	User Row ID	Standard Order	Block Number	Replicate Number	Point Type	A	B	C	Y1	Y2
1		1	Block 1	Replicate 1	Design Point	0.4	0.55	0.05	24.26	7.32
2		2	Block 1	Replicate 1	Design Point	0.44	0.55	0.01	35.46	19.88
3		3	Block 1	Replicate 1	Design Point	0.4	0.52	0.08	28.73	17.24
4		4	Block 1	Replicate 1	Design Point	0.75	0.24	0.01	20	12.63
5		5	Block 1	Replicate 1	Design Point	0.75	0.2	0.05	40.99	21.58
6		6	Block 1	Replicate 1	Design Point	0.72	0.2	0.08	20.87	12.4
7		7	Block 1	Replicate 1	Design Point	0.5767	0.3767	0.0467	39.33	12.76
8		8	Block 1	Replicate 1	Design Point	0.4883	0.4633	0.0483	40.36	13.33
9		9	Block 1	Replicate 1	Design Point	0.5083	0.4633	0.0283	30.55	11.56
10		10	Block 1	Replicate 1	Design Point	0.4883	0.4483	0.0633	45.12	21.12
11		11	Block 1	Replicate 1	Design Point	0.6633	0.3083	0.0283	25.23	15.93
12		12	Block 1	Replicate 1	Design Point	0.6633	0.2883	0.0483	34.22	21.34
13		13	Block 1	Replicate 1	Design Point	0.6483	0.2883	0.0633	39.85	24.67
14										



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Select Column(s)

☒

Select Columns Manually

☐

Select Matching Columns from Node

☐

Select Columns Manually and from Node

Select Node

Excluded Columns

Col2 -- Standard Order

Col3 -- Block Number

Col4 -- Replicate Number

Col5 -- Point Type

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>

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Included Columns

Col6 -- A

Col7 -- B

Col8 -- C

Col9 -- Y1

Col10 -- Y2

Execute

Cancel

Results :

Output							
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	
User Header	User Row ID	A	B	C	Y1	Y2	
1		0.4	0.55	0.0500000	24.26	7.32	
2		0.4400000	0.55	0.01	35.46	19.88	
3		0.4	0.52	0.08	28.73	17.24	
4		0.75	0.24	0.01	20	12.63	
5		0.75	0.2	0.0500000	40.99	21.58	
6		0.72	0.2	0.08	20.87	12.4	
7		0.5766667	0.3766667	0.0466667	39.33	12.76	
8		0.4883333	0.4633333	0.0483333	40.36	13.33	
9		0.5083333	0.4633333	0.0283333	30.55	11.56	
10		0.4883333	0.4483333	0.0633333	45.12	21.12	
11		0.6633333	0.3083333	0.0283333	25.23	15.93	
12		0.6633333	0.2883333	0.0483333	34.22	21.34	
13		0.6483333	0.2883333	0.0633333	39.85	24.67	

Step 4: Mixture Analysis – Y1 – Full Cubic

Create a new tab named “Regression-Y1-Full-Cubic”. Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

From DOE → Post DOE Analysis → Mixture Analysis, select Y1 as the response, A, B, and C as the mixture components, and choose Full Cubic as the analysis type. Execute the analysis to obtain the fitted mixture model and the corresponding statistical results.

Input						
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)
User Header	User Row ID	A	B	C	Y1	Y2
1		0.4	0.55	0.05	24.26	7.32
2		0.44	0.55	0.01	35.46	19.88
3		0.4	0.52	0.08	28.73	17.24
4		0.75	0.24	0.01	20	12.63
5		0.75	0.2	0.05	40.99	21.58
6		0.72	0.2	0.08	20.87	12.4
7		0.5767	0.3767	0.0467	39.33	12.76
8		0.4883	0.4633	0.0483	40.36	13.33
9		0.5083	0.4633	0.0283	30.55	11.56
10		0.4883	0.4483	0.0633	45.12	21.12
11		0.6633	0.3083	0.0283	25.23	15.93
12		0.6633	0.2883	0.0483	34.22	21.34
13		0.6483	0.2883	0.0633	39.85	24.67

Input							Action
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	
User Header	User Row ID	A	B	C	Y1	Y2	
1		0.4	0.55	0.0500000	24.26	7.32	
2		0.4400000	0.55	0.01	35.46	19.88	
3		0.4	0.52	0.08	28.73	17.24	
4		0.75	0.24	0.01	20	12.63	
5		0.75	0.2	0.0500000	40.99	21.58	
6		0.72	0.2	0.08	20.87	12.4	
7		0.5766667	0.3766667	0.0466667	39.33	12.76	
8		0.4883333	0.4633333	0.048			
9		0.5083333	0.4633333	0.028			
10		0.4883333	0.4483333	0.063			
11		0.6633333	0.3083333	0.028			
12		0.6633333	0.2883333	0.048			
13		0.6483333	0.2883333	0.063			
14							
15							
16							

Show Spreadsheet Toolbar
Import from File
Import from Database
Import from Spreadsheet
Re-Import from Spreadsheet
Import from Multiple Spreadsheets
Adjust Spreadsheet Precision
Export Spreadsheet Data

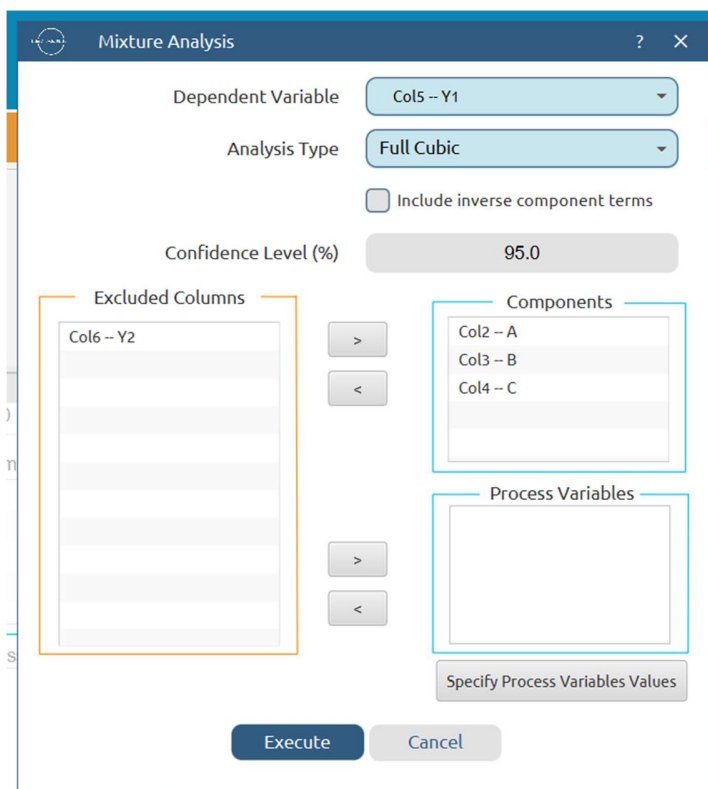
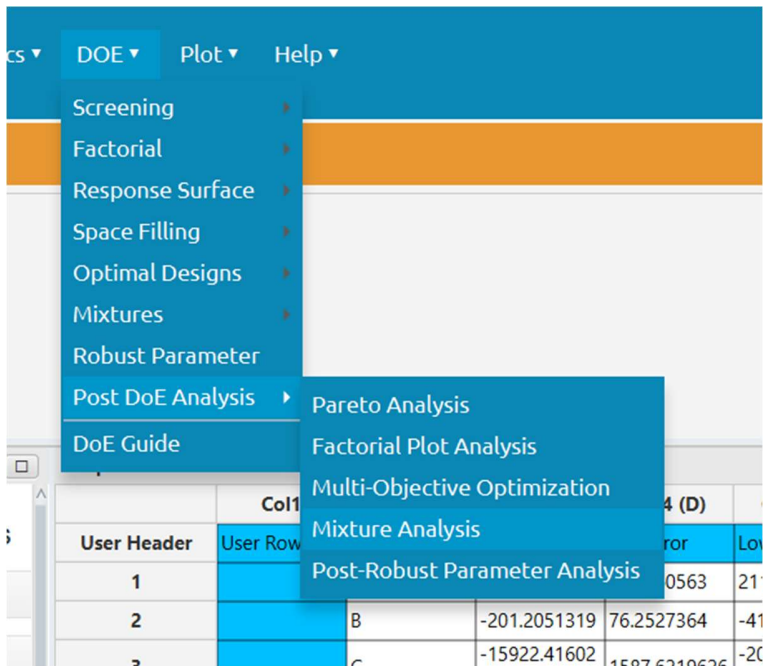
Choose tab as input

Select input tab

Data

Execute

Cancel



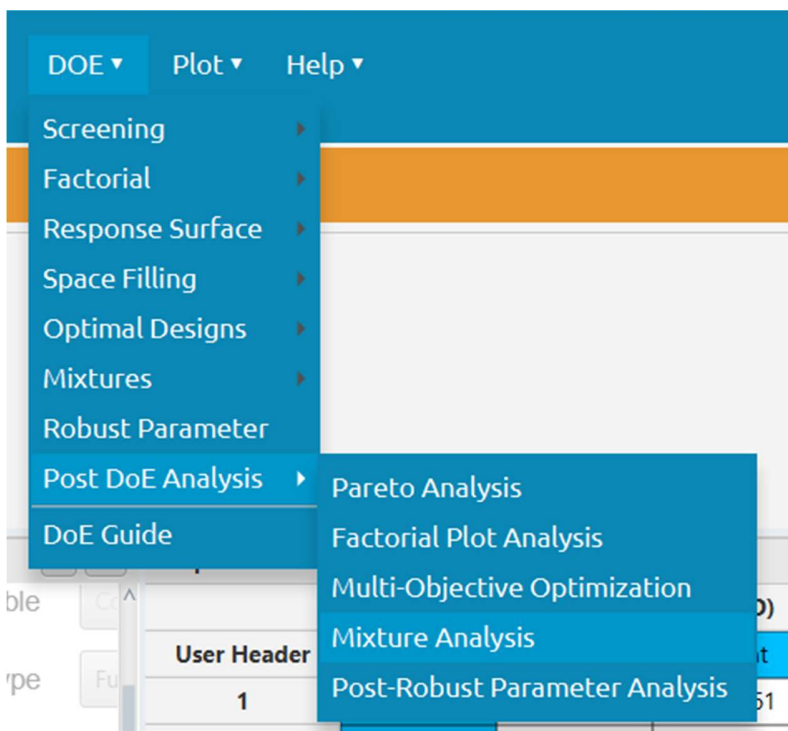
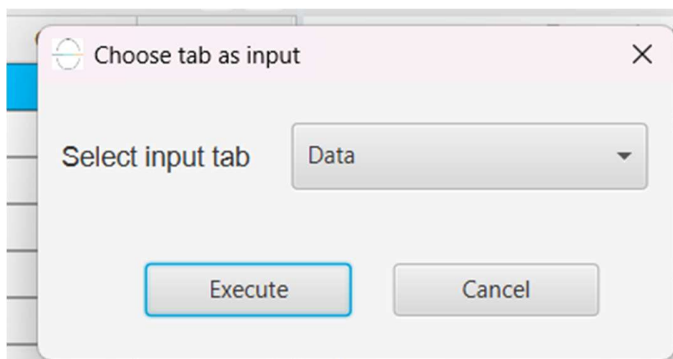
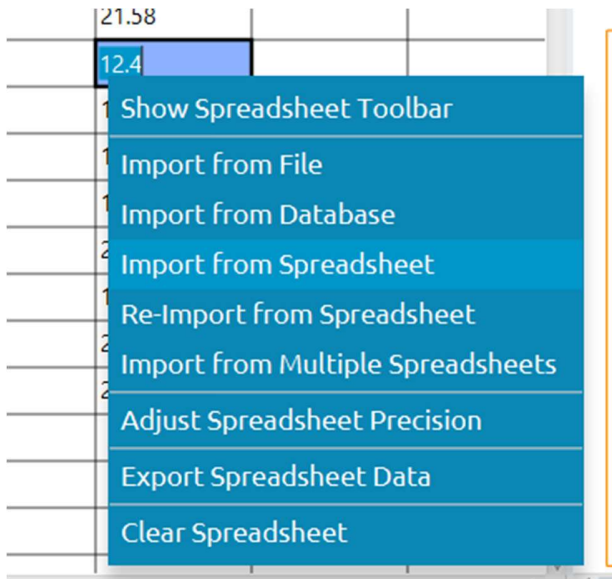
Results :

Output										
	Col1	Col2 (S)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)
User Header	User Row ID	Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	p-value	VIF
1		A	277.2357489	23.6360563	211.6115360	342.8599618	11.7293573	1.0	0.0003022	1607.6032960
2		B	-201.2051319	76.2527364	-412.9166685	10.5064047	-2.6386611	1.0	0.0576611	7567.0383017
3		C	-15922.4160275	1587.6219626	-20330.3612543	-11514.4708007	-10.0290979	1.0	0.0005557	55261.1952889
4		A*B	-55.4006409	145.6076450	-459.6722742	348.8709924	-0.3804789	1.0	0.7229322	7293.8118669
5		A*C	13121.1334866	2671.5630420	5703.6853557	20538.5816174	4.9114070	1.0	0.0079784	53264.7038660
6		B*C	31315.7811600	1899.3728240	26042.2767803	36589.2855397	16.4874325	1.0	0.0000792	12094.1135096
7		A*B*(A-B)	-1491.2596307	166.1056894	-1952.4429589	-1030.0763024	-8.9777758	1.0	0.0008519	707.3178940
8		A*C*(A-C)	1272.5060559	1607.8278810	-3191.5397938	5736.5519057	0.7914442	1.0	0.4729711	6372.1396718
9		B*C*(B-C)	-25409.0993530	1607.8278810	-29873.1452028	-20945.0535032	-15.8033703	1.0	0.0000937	1389.6402923
10										

Step 5: Mixture Analysis – Y2 – Full Cubic

Repeat the previous step for Y2. Create a tab named “Regression-Y2-Full Cubic”, import A, B, C, Y1 and Y2 from the Data tab, and apply Mixture Analysis with the settings below.

	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)
r	User Row ID	A	B	C	Y1	Y2
		0.4	0.55	0.0500000	24.26	7.32
		0.4400000	0.55	0.01	35.46	19.88
		0.4	0.52	0.08	28.73	17.24
		0.75	0.24	0.01	20	12.63
		0.75	0.2	0.0500000	40.99	21.58
		0.72	0.2	0.08	20.87	12.4
		0.5766667	0.3766667	0.0466667	39.33	12.76
		0.4883333	0.4633333	0.0483333	40.36	13.33
		0.5083333	0.4633333	0.0283333	30.55	11.56
		0.4883333	0.4483333	0.0633333	45.12	21.12
		0.6633333	0.3083333	0.0283333	25.23	15.93
		0.6633333	0.2883333	0.0483333	34.22	21.34
		0.6483333	0.2883333	0.0633333	39.85	24.67



Mixture Analysis

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Dependent Variable

Col6 -- Y2

Analysis Type

Full Cubic

☐ Include inverse component terms

Confidence Level (%)

95.0

Excluded Columns

Col5 -- Y1

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Components

Col2 -- A

Col3 -- B

Col4 -- C

Process Variables

Specify Process Variables Values

Execute

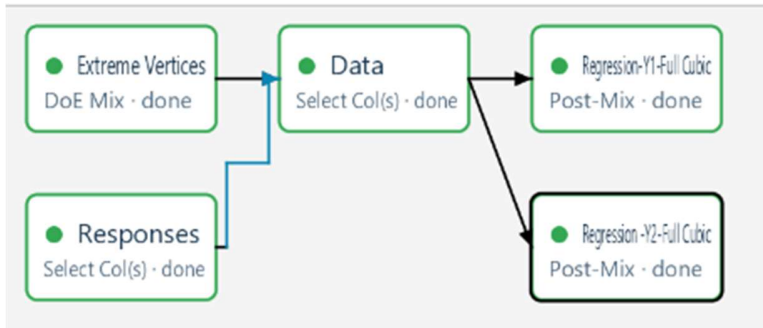
Cancel

Results :

Output											
	Col1	Col2 (S)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)	
User Header	User Row ID	Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	p-value	VIF	
1		A	69.6497061	52.8545463	-77.0980403	216.3974526	1.3177619	1.0	0.2579876	1607.6032960	
2		B	258.9759765	170.5150696	-214.4497538	732.4017067	1.5187864	1.0	0.2034348	7567.0383017	
3		C	-3857.4009526	3550.2131777	-13714.3729515	5999.5710462	-1.0865266	1.0	0.3383400	55261.1952889	
4		A*B	-593.5381350	325.6053345	-1497.5634720	310.4872021	-1.8228760	1.0	0.1424006	7293.8118669	
5		A*C	-1843.4186262	5974.1037476	-18430.1897331	14743.3524808	-0.3085682	1.0	0.7730526	53264.7038660	
6		B*C	15682.6563476	4247.3451411	3890.1357213	27475.1769739	3.6923433	1.0	0.0209756	12094.1135096	
7		A*B*(A-B)	16.3348112	371.4427120	-1014.9554883	1047.6251107	0.0439767	1.0	0.9670308	707.3178940	
8		A*C*(A-C)	5334.2997333	3595.3973078	-4648.1235227	15316.7229892	1.4836468	1.0	0.2120586	6372.1396718	
9		B*C*(B-C)	-19689.7487421	3595.3973079	-29672.1719982	-9707.3254860	-5.4763763	1.0	0.0054115	1389.6402923	

Final Isalos Workflow

The final workflow is presented below :



References

Damiri, S.; Pouretedal, H. R.; Bakhshi, O. An extreme vertices mixture design approach to the optimization of methylal production process using p-toluenesulfonic acid as catalyst. Chemical Engineering Research and Design 2016, 112, 155–162.